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ABOUT ME

Adaptable professional with a quick-learning ability and a talent for adjusting to new environments. Skilled in rapidly acquiring new knowledge and applying it effectively. Driven by a passion for continuous learning and successfully navigating change.

WORK EXPERIENCE

🏢 **Nook Office Solutions** – Islamabad, Pakistan

Junior GIS Analyst

[20/10/2025 – Current]

- Translated intricate **urban planning codes** into structured spatial data to support **large-scale zoning analysis and mapping** projects.
- Systematically developed and maintained datasets for land-use classifications and dimension regulations, successfully integrating **qualitative policy data** into GIS shapefiles.
- Engineered custom **Python-based tools** and scripts to **automate** data cleaning, batch file creation, and cloud upload processes, significantly reducing manual processing time.
- Contributed to the iterative refinement of internal QA tools, improving the accuracy and validation speed of geospatial deliverables.

🏢 **National Disaster Risk Management Fund** – Islamabad, Pakistan

GIS Intern

[12/06/2025 – 13/09/2025]

- Digitized and managed geospatial datasets on Pakistan's **energy infrastructure** to support national disaster risk assessments.
- Conducted **hazard assessments** for multiple disaster scenarios (floods, earthquakes, and other natural hazards) to evaluate infrastructure vulnerability.
- Conducted **flood extent assessment** of the recent floods in Punjab using **Sentinel-1 SAR imagery**, comparing observed inundation with **flood hazard return periods** across river basins to evaluate the magnitude and rarity of the event.
- Generated **flood damage maps (2022)** to highlight the social, economic, and infrastructural impacts of the catastrophic floods in Pakistan.
- Carried out extensive **geospatial analysis of the Galiyat region** for risk mapping and spatial planning support.
- Researched and explored data sources and methodologies for **Urban Heat Island (UHI)** analysis.
- Developed **Python-based scripts** for automated acquisition of satellite data and integrated outputs into a **PostgreSQL/PostGIS database**.
- Received hands-on training in **React.js** for developing interactive geospatial web applications.

🏢 **Global Climate-Change Impact Studies Centre** – Islamabad, Pakistan

Research Intern

[07/2023 – 08/2024]

- Conducted **in-depth research** on the impacts of **climate change** on various ecosystems and communities.
- Focused on the **development of RUSLE based tool** for analyzing climate induced soil erosion and reservoir sedimentation in the Tarbela Region.
- Worked on mangrove restoration and suitability analysis using multiple **MCDM techniques**.
- Developed **Python-based toolboxes** to streamline analyses and geoprocessing tasks
- Analyzed data, and contributed to comprehensive reports.
- Assisted in the organization of **International Conference** on the Official Launch of the Transparency Platform for Pakistan (**RISQ**).
- Assisted in the organization of research-related events, such as workshops and seminars.

 **Gravity Solutions** – Rawalpindi, Pakistan

Cadastral Mapping

[07/2024 – 09/2024]

- Digitized different types of land parcels.
- Ensured accuracy by performing quality checks.
- Developed and maintained databases.
- Maintained fields and records.
- Removed topological errors.
- Developed Python-based scripts to streamline processes.

 **TerraCarta Technologies** – Rawalpindi, Pakistan

GIS Analyst

[08/2023 – Current]

- Occupation Overview: Freelancing.
- Managed multiple freelancing projects, ensuring timely delivery, and client satisfaction.
- Edited and enhanced on-demand GIS data to meet diverse project requirements.
- Developed and maintained GIS databases, ensuring data accuracy and consistency for various projects.
- Utilized scripting languages such as Python or VBScript to automate GIS processes and tasks.
- Created maps and visualizations of geographic information systems data to support different objectives.
- Created detailed reports based on observations from performing spatial analysis.

EDUCATION AND TRAINING

Master's in Remote Sensing & GIS

National University of Science & Technology [2025 – Current]

City: Islamabad | Country: Pakistan | Website: igis.nust.edu.pk

BS Geoinformatics

PMAS Arid Agriculture University [2021 – 2025]

City: Rawalpindi | Country: Pakistan | Website: www.uaar.edu.pk/ | Field(s) of study: Remote Sensing & GIS |
Final grade: 3.61 | Thesis: Developing a Web-Based Tool for Spatio-Temporal Crop Mapping using Sentinel Imagery in Punjab Region

Intermediate

Punjab College Ayub Park Campus [08/2018 – 04/2020]

City: Rawalpindi | Country: Pakistan

Matriculation

Army Public School Westridge III [03/2016 – 03/2018]

City: Rawalpindi | Country: Pakistan

PUBLICATIONS

[2026]

Developing a RUSLE-Based Tool for Analyzing Climate Induced Soil Erosion and Reservoir Sedimentation in the Tarbela Region

Authors: Muhammad Arif Goheer, Sher Shah Hassan, Muhammad Taimoor Ashfaq Malik, Syeda Rosheen Raza Kazmi, Syed Zee Ali Shan | Journal Name: International Journal of Environmental Research | Publisher: Springer Nature

PROJECTS

[02/03/2026 – 07/04/2026]

Developing AHP/FAHP Spatial Analyzer Add-in for ArcGIS Pro

- **Developed** a professional-grade ArcGIS Pro Add-in (**C#, .NET 8.0**) for Multi-Criteria Decision Analysis (MCDA), implementing both Analytic Hierarchy Process (AHP) and Fuzzy AHP (FAHP) methodologies.
- **Engineered** a custom **WPF user interface** featuring a dynamic pairwise comparison matrix with real-time consistency ratio (CR) validation and automated reciprocal calculations.
- **Integrated** a **Python-based** spatial engine (ArcPy) to perform high-resolution weighted raster overlays and automated suitability mapping directly within the GIS environment.
- **Implemented** advanced analytical features, including a Reclassification Manager for raster statistics and a Monte Carlo Sensitivity Analysis module to test the robustness of spatial decisions.
- **Ensured** cross-version compatibility for ArcGIS Pro 3.5 and 3.6+ using optimized NuGet-based SDK management and standard MSBuild deployment.
- **Key Technologies:** C#, .NET 8.0/WPF, Python (ArcPy), ArcGIS Pro SDK, Git/GitHub.

Link: <https://github.com/tcantbenormal/Ahp-Fahp-Analyzer-For-ArcGIS-Pro?tab=readme-ov-file>

Developing a GeoAI-Driven Spatial Decision Support System for Rainwater Harvesting Suitability Zones Identification (Aqua-Assist)

- Architected "Aqua-Assist," a flagship GeoAI-Driven Spatial Decision Support System (SDSS) to dynamically evaluate Rainwater Harvesting (RWH) site suitability zones across KPK.
- Engineered a custom Fuzzy Analytical Hierarchy Process (FAHP) from scratch using Python libraries to replace static GIS weight matrices with dynamic, localized parameter prioritization to resolve complex multi-criteria spatial decisions.
- Pioneered the integration of an offline Large Language Model (Ollama) and Explainable GeoAI utilizing the LangChain framework to empower the system to autonomously adjust spatial weights based on specific terrain contexts while providing highly defensible geographic justifications to stakeholders.
- Developed a robust dual-pipeline data ingestion framework capable of extracting and processing critical hydrological and terrain parameters directly from raw DEMs and multispectral satellite rasters.
- Delivered a fully functional, offline desktop application featuring an interactive multiplex GUI built with streamlit and folium and bridged the gap between complex spatial mathematics and practical, human-centric agricultural water management.

Building a National Spatial Data Infrastructure (NSDI) for Pakistan: Road Network Digitization

- Developed a refined road network dataset as a foundational component of Pakistan's NSDI, addressing fragmentation, incomplete metadata, and misaligned geometries.
- Aimed to create a seamless, standardized geospatial database by translating Urdu road names to English, enriching attributes, and aligning features with high-resolution basemaps.
- Reduced road segment redundancies across regions (e.g., Punjab: 23,000 to 2,131 segments; Sindh: 15,000 to 3,000; KPK: 3,000 to 1,000) through data merging and spatial adjustments using QGIS and ArcGIS Pro.
- Conducted topology checks and quality assurance, cross-referencing with Google Roads basemaps and stakeholder reviews to ensure accuracy and reliability.

- Produced a streamlined, navigable dataset supporting geospatial analysis, enhancing data-sharing capabilities for decision-making in sustainable development.
- Demonstrated expertise in spatial data management, OpenStreetMap data extraction, and geospatial tools, contributing to a scalable model for SDI development.

Disaster Management Plan for Glacial Lake Outburst Flood (GLOF) in Shishper Glacier, Pakistan

- Developed a comprehensive disaster management plan to address GLOF risks at Shishper Glacier, Hunza Valley, using GIS and remote sensing to analyze the May 2022 outburst event.
- Aimed to investigate the breach causes, assess downstream impacts, and devise strategies for preparedness, response, recovery, and mitigation to safeguard communities.
- Utilized Landsat-8/9, Sentinel-2, MODIS, and ALOS PALSAR data to calculate indices and map changes in snow cover, debris, turbidity, and land surface temperature, identifying rapid glacier melt and debris expansion as key GLOF triggers.
- Quantified post-flood impacts, including increased stream turbidity and infrastructure damage (e.g., Hassanabad Bridge collapse), using satellite-derived indices.
- Proposed a disaster management framework with early warning systems, community training, lake drainage control, and climate-resilient infrastructure to mitigate future GLOF risks.
- Demonstrated proficiency in geospatial analysis, hazard mapping, and remote sensing, enhancing resilience strategies for climate-driven hazards in the Karakoram region.

Developing a RUSLE-Based Tool for Analyzing Climate Induced Soil Erosion and Reservoir Sedimentation in the Tarbela Region Designed and implemented a Python-based GIS tool to model soil erosion and sedimentation in the Tarbela Reservoir using the Revised Universal Soil Loss Equation (RUSLE). The project integrated satellite imagery, climate datasets, and soil property data to quantify erosion trends and assess reservoir storage loss over three decades. Results revealed a **65% reduction in reservoir capacity** due to increased sediment deposition, providing crucial insights for water management, hydropower planning, and climate adaptation strategies in Pakistan.

Identification of Potential Rainwater Harvesting Zones and Their Evaluation for Agricultural Production in Selected Regions of Khyber Pakhtunkhwa, Pakistan

- Developed and implemented a comprehensive geospatial framework to identify optimal rainwater harvesting (RWH) zones in selected districts of Khyber Pakhtunkhwa to enhance agricultural productivity and address water scarcity.
- Utilized advanced tools including ArcGIS, QGIS, and Google Earth Engine to integrate datasets for multi-criteria decision analysis.
- Applied Fuzzy Analytical Hierarchy Process (FAHP) to rank RWH sites based on topography, soil, rainfall, and aridity indices, achieving precise site suitability mapping.
- Outcomes included high-resolution RWH hotspot maps, a master plan for policy-makers, and evidence-based strategies for sustainable water management, demonstrating enhanced land and water productivity in arid and semi-arid regions.

Mapping Salt-Affected Soils Using Remote Sensing and GIS (Chakwal, Pakistan)

- Conducted a comprehensive study to identify and map salt-affected soils in Chakwal district using satellite-based spectral indices (e.g., NDSI, NDVI, VSSI, SI1–SI4, Salinity Ratio, MSAVI, Brightness Index) from Landsat 8/9 and Sentinel-2 imagery.
- Aimed to assess soil salinity distribution, evaluate the effectiveness of salinity and vegetation indices across seasons, and classify land cover to support agricultural management.
- Developed a salinity zonation map through unsupervised classification, categorizing areas into low (15,222.26 ha), moderate (112.73 ha), and high (209.10 ha) salinity levels, validated using high-resolution Google Earth imagery.
- Quantified salinity distribution across land use types, revealing significant salinization impacting agricultural productivity in the Indus Basin.

- Provided actionable insights for reclamation strategies, including salt-tolerant crops and improved irrigation, to enhance food security and sustainable farming practices.
- Utilized ArcGIS Pro for data processing, including multiband raster stacking, masking, and zonal statistics, demonstrating proficiency in geospatial analysis and remote sensing techniques.

Developing Shapefile to KML Converter Desktop Application

- Developed a Python-based desktop application to convert shapefiles to KML format, enhancing accessibility for visualization in platforms like Google Earth.
- Aimed to streamline spatial data conversion by enabling users to load shapefiles from ZIP archives, reproject coordinates to WGS84, and generate KML files through a user-friendly GUI.
- Utilized Python libraries (tkinter, PyProj, SimpleKML, PIL, Shapefile, ZipFile) to build a robust tool for file handling, coordinate transformation, and KML generation.
- Designed an intuitive tkinter-based interface with dynamic background resizing, file selection dialogs, and error handling, ensuring usability for users with varying GIS expertise.
- Packaged the application into a standalone executable using PyInstaller, incorporating custom icons and additional files for seamless distribution.
- Demonstrated proficiency in Python programming, GIS data processing, and GUI development, delivering a practical tool for geospatial data management.

SKILLS

ArcGIS Pro / QGIS / Erdas Imagine / ENVI / Trimble eCognition / Google Earth Engine / JavaScript / Node.js, React.js / ETL Operations / Databases (PostgreSQL/PostGIS, MySQL, Redis) / Python Django Web Frameworks / OSGeo4W(QGIS, GRASS GIS, SAGA GIS) / Python / LLM Integration & LangChain / GDAL / Spatial Decision Support Systems / AutoCAD: Intermediate / Visualization (Jupyter, d3.js, OpenLayers.js, Leaflet.js) / C# / .NET Development / ArcGIS Pro SDK for .NET

CONFERENCES AND SEMINARS

[05/11/2024 – 05/11/2024] PMAS Arid Agriculture University

Upscaling Irrigation and Climate Advisory Services Through Citizen Science

- 1-day seminar on good irrigation practices and their long term effects on climate.
- Initiative aimed at empowering farmers with real-time climate insights and smart irrigation solutions for sustainable agriculture.

[08/10/2024 – 08/10/2024] Best Western Hotel, Islamabad

Third National Communication (TNC) to UNFCCC

- **National Consultative Workshop** on "Research and Systematic Observation"
- Chapter of Pakistan's **Third National Communication (TNC) to UNFCCC**

[04/09/2003 – 07/09/2003] Islamabad, Pakistan

Training on National GHG Inventory Management and MRV System

- Organized by the Global Climate Change Impact Studies Centre (GCISC), Ministry of Climate Change, and Environmental Coordination under the Third National Communication on Climate Change Project.
- Aimed at enhancing understanding of MRV systems crucial for tracking greenhouse gas emissions and informing climate policy decisions.
- Discussed the country's rising temperatures, increased frequency of heatwaves, and erratic rainfall patterns, underscoring the need for robust climate action

[19/09/2023 – 21/09/2023] Islamabad, Pakistan

Training on Transparency Platform for Pakistan

- Organized by the Global Climate Change Impact Studies Centre (GCISC), Ministry of Climate Change and Environmental Coordination (MoCC&EC), and CITEPA France, with financial support from GIZ Pakistan.
- Launched the RISQ tool, developed collaboratively by CITEPA France and GCISC, aimed at enhancing Measurement, Reporting, and Verification (MRV) for tracking mitigation and adaptation actions.
- Emphasized developing frameworks for compiling GHG inventories, data collection, sharing, and reporting, aligning with UNFCCC guidelines and the Paris Agreement.
- Highlighted the importance of monitoring and evaluation systems for adaptation tracking, particularly in the agriculture and water sectors.

[17/01/2025 – 17/01/2025] PMAS Arid Agriculture University, Rawalpindi

1 Day Training Seminar on AutoCAD Civil 3D

- Organized by the Department of Land & Water Conservation Engineering, PMAS-AAUR.
- Focused on introducing the fundamentals of AutoCAD Civil 3D, covering essential tools and workflows for civil infrastructure design, including surface modeling, alignments, and grading techniques.

CERTIFICATIONS

Introduction to GIS Mapping

University of Toronto | Coursera

Link: https://drive.google.com/file/d/1QQFBG7BORyUfvZZ0ySYHMRp8w1-8amvq/view?usp=drive_link

Fundamentals of GIS

UC Davis | Coursera

Link: <https://drive.google.com/file/d/1SlejTx9WtmqZAL6fQ-Gr2FQwyHF78sE/view?usp=sharing>

Going Places With Spatial Analysis

ESRI MOOC

Link: https://drive.google.com/file/d/13ES7v01DLFHZfuc7V5Em5ur0TqGy15td/view?usp=drive_link

Make An Impact with Modern GeoApps

ESRI MOOC

Link: https://drive.google.com/file/d/1UW4I7FyjrURtkQ4UJDFHI-kTczuzSIK/view?usp=drive_link

Spatial Data Science: The New Frontier in Analytics

ESRI MOOC

Link: https://drive.google.com/file/d/1yhj7vZl7AhvfBxAj4f9EsmurK8vHNh2d/view?usp=drive_link

Basics of AUTOCAD Civil 3D

Department of Land & Water Conversation Engineering | PMAS Arid Agriculture University Rawalpindi

Link: https://drive.google.com/file/d/1hZjRfWxEchw2Cgfsi5Zcum1Y8TZMy-3/view?usp=drive_link

LANGUAGE SKILLS

Mother tongue(s): Urdu

Other language(s):

English

LISTENING C1 READING C1 WRITING C1

SPOKEN PRODUCTION C1 SPOKEN INTERACTION C1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user